

EXPERIMENT No. 1

Date: 03-02

DETERMINATION OF THE COEFFICIENT OF VISCOSITY OF LIQUID BY STOKES'S METHOD BY GRAPHICALLY ANALYZING THE VARIATION OF TIME TAKEN FOR SIX METAL BALLS OF DIFFERENT DIAMETERS TO TRAVEL THE SAME DISTANCE IN THE GIVEN LIQUID WITH RESPECT TO THEIR DIAMETERS.

APPARATUS REQUIRED:

- Measuring cylinder
- Steel ball bearing or glass beads of varying diameter
- Experimental liquid
- Micrometer screw gauge
- Stop watch
- Meter scale
- Thermometer 0 – 100°C

THEORY:

When a small steel ball is allowed to fall through a viscous liquid, after a certain time it will attain a terminal velocity v_t such that

$$\text{i.e. } 6\pi\eta r v_t = \frac{4}{3}\pi r^3 (\rho - \sigma) g$$

$$\therefore \eta = \frac{2}{9} g (\rho - \sigma) \times \frac{r^2}{v_t} \dots\dots\dots (1)$$

In terms of diameter

$$\eta = \frac{1}{18} d^2 \frac{(\rho - \sigma) g}{v_t} \quad \text{since } v_t = \frac{x}{t}$$

$$\therefore \eta = \frac{1}{18} d^2 \frac{(\rho - \sigma) g \times t}{x} \dots\dots\dots (2)$$

Where η is the coeff. of viscosity

ρ is the density of the ball

σ is the density of the liquid

r is the radius of the ball

v_t is the density of the terminal velocity of the ball

If we plot a graph between time and diameter

of a curve of the form $y \propto \frac{1}{x^2}$ is obtained.

For angle 45°, $t = d^2$ and knowing the value of ρ , σ and g , the value of coefficient of viscosity η can be determined.

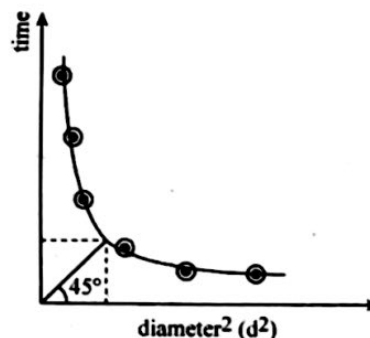


Fig. 3.3: Graph of t vs d^2

OBSERVATIONS:Zero instrumental error of micrometer screw gauge = -0.02 mm (+ve or -ve)Zero error correction = $+0.02$ mmDensity of Glycerine, (σ) = 1.26 g cm $^{-3}$ Density of steel (ρ) = 7.7 g cm $^{-3}$ Temp. of glycerine = 20 °CL.C. of scale of viscometer = 0.1 cm**Table 1: Diameter of the balls**

Ball	Reading of the			Total (a+c) mm	Apparent mean diameter mm	Corrected mean diameter d mm	Corrected mean diameter d cm
	Main scale (a) mm	Circular Scale (b)	Value of circular Scale reading $c = (b \times LC)$ mm				
Smallest 1	3	1	0.01	3.01	3.02	3.04	0.304
	3	3	0.03	3.03			
2	3	8	0.08	3.08	3.05	3.07	0.305
	3	10	0.10	3.10			
3	3	2	0.02	3.02	3.03	3.05	0.307
	3	4	0.04	3.04			
4	4	56	0.56	4.56	4.57	4.59	0.459
	4	58	0.58	4.58			
5	4	51	0.51	4.51	4.54	4.56	0.456
	4	57	0.57	4.57			
Largest 6	4	66	0.66	4.66	4.67	4.69	0.466
	4	68	0.68	4.68			

Table 2: Time of descent and terminal velocity of the ball

Spherical ball	Distance between A & B (x) cm	Time of fall from A to B (t) s
Smallest 1	50	11.84
2	50	18.94
3	50	17.07
4	50	9.17
5	50	8.66
Largest 6	50	8.47

The coefficient of viscosity of the liquid by stocks method,

$$\therefore \eta_1 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.304)^2 (7.7 - 1.26) \times 980 \times 18.34}{50}$$

$$11.88 \text{ dyne s cm}^{-2}$$

$$\therefore \eta_2 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.305)^2 (7.7 - 1.26) \times 980 \times 18.94}{50}$$

$$12.27 \text{ dyne s cm}^{-2}$$

$$\therefore \eta_3 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.307)^2 (7.7 - 1.26) \times 980 \times 17.07}{50}$$

$$11.06 \text{ dyne s cm}^{-2}$$

$$\therefore \eta_4 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.459)^2 (7.7 - 1.26) \times 980 \times 9.17}{50}$$

$$13.54 \text{ dyne s cm}^{-2}$$

$$\therefore \eta_5 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.456)^2 (7.7 - 1.26) \times 980 \times 8.66}{50}$$

$$12.79 \text{ dyne s cm}^{-2}$$

$$\therefore \eta_6 = \frac{1}{18} d^2 \frac{(\rho - \sigma) g x t}{x} = \frac{1}{18} \times \frac{(0.466)^2 (7.7 - 1.26) \times 980 \times 8.47}{50}$$

$$12.89 \text{ dyne s cm}^{-2}$$

$$\text{Now, the mean } \eta = \frac{\eta_1 + \eta_2 + \eta_3 + \eta_4 + \eta_5 + \eta_6}{6} = 12.45 \text{ dyne s cm}^{-2}$$

RESULT:

From the experiment coeff. of viscosity of given liquid is found to be $12.45 \text{ dyne cm}^{-2}$.

CONCLUSION:

Hence, the coefficient of viscosity of liquid is found using Stokes's method.

SOURCES OF ERROR:

- 1) The presence of impurities in the liquid & on the surface of ball.
- 2) The metallic balls not being perfectly spherical.
- 3) Starting of stopwatch without gaining terminal velocity by the balls.
- 4) Use of larger sized steel ball.

PRECAUTIONS:

- 1) The ball should be released from the center of jar so that they don't touch the wall of jar.
- 2) Avoid contaminating the balls, use tweezers or tissue paper to hold balls.
- 3) Steel balls should be small.
- 4) The upper calibration point p is chosen so that the balls gain terminal velocity at that point.

VIVA QUESTIONS:

1. What is Stoke's law? Explain.
2. Define terminal velocity. How is it related with the radius of the ball?
3. Define viscosity and coefficient of viscosity.
4. State equation of continuity.
5. What is the principle on which Bernoulli's theorem is based?
6. Does the coefficient of viscosity depends upon temperature?
7. Why cloud appears to be floating in the sky?
8. What will happen if the density of the medium is greater than that of the body?
9. What do you mean by velocity gradient and pressure gradient?
10. What are streamline and turbulent flow?

1) If a sphere of radius ' r ' moves through fluid of coeff. of viscosity (η) with a terminal velocity ' u ' then viscous force acting on sphere is $F = 6\pi\eta r u$

2) The constant velocity attained by body when moves through viscous liquid in which resultant force acting on it is zero. which is called terminal velocity. It is inversely proportional to radius of ball.

3) Viscosity is defined as the property of liquid by virtue of which it oppose the motion between layers.

Coef. of viscosity is defined as tangential force acting per unit area per unit velocity gradient normal to direction of flow of liquid.

4) For steady flow of incompressible and non-viscous fluid through tube of non-uniform CSA of

every point remain constant. i.e. $Av = \text{constant}$.

5) Bernoulli's theorem is based on the principle of conservation of energy.

6) Yes, coeff. of viscosity (η) $\propto \frac{1}{\text{temperature}}$.

7) The cloud floats in air due to viscosity of air.

8) If density of medium is greater than body then the body will float on it.

9) Whenever a layer of liquid moves over the next layer of liquid surface the variation of its relative velocity 'v' with its distance between layer is called velocity gradient. Thus, $\frac{dv}{dx}$ is velocity gradient.

city gradient.

When a liquid flow through a capillary tube in stream line, motion under pressure applied to one end, then variation of p & l (length of tube) is pressure gradient.

10) The flow of liquid in which every particle of liquid flows exactly the path of its proceeding and has same velocity as that of its proc-

eeding particle is called streamline flow.

The flow of liquid in which each particle doesn't have a definite path so that paths of individual particle cross each other is called turbulent flow.

103-16